Pennine Services

Operational Notes

The starting sessions have sufficient routes preset to allow a few minutes of operation without any operator actions being necessary. Not all services have been included, the simulations focus on those that cross the Pennines.

Freight services have very generous time allowances and should be held back whenever there may be conflicts with passenger services.

Bradford Interchange platform 4 should be used for all services departing towards New Pudsey.

Check Leeds exit times in determining whether services from Brighouse or from Huddersfield should have priority in going forward towards Leeds.

Services should use platforms 2 or 4 at Blackburn unless the description says otherwise.

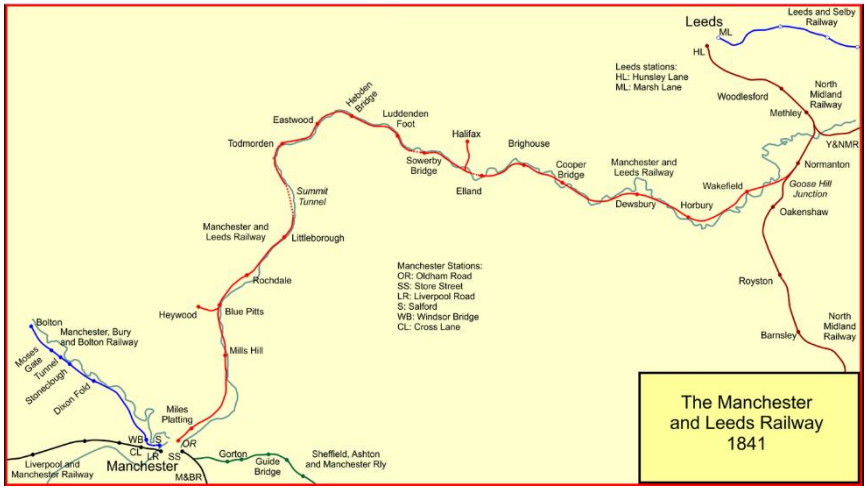
Trains from Barnsley should use the terminal platform at Huddersfield.

Note that the track element at 59-35 is a bridge so there is no need to hold trains there.

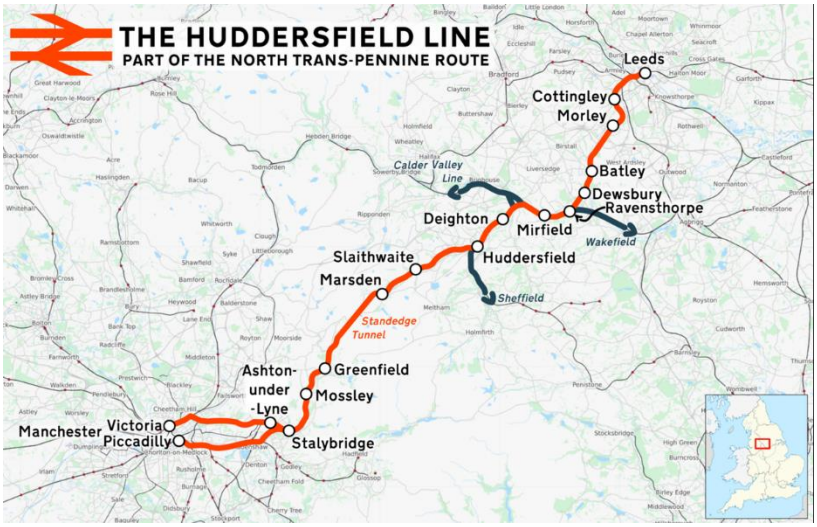
After 6D74 has exited UPL an automatic route can be set across that loop as no other trains use it.

A Brief History of the Routes

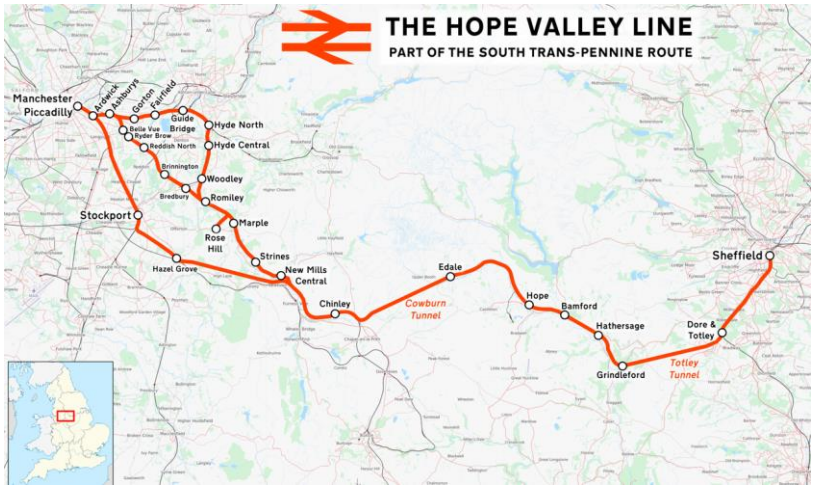
There are four major railway routes across the Pennines. The first of these to be built was the Manchester and Leeds Railway via Todmorden, authorised in 1836 from Manchester to Normanton where it joined the North Midland Railway with running powers from there into Leeds. The engineer was George Stephenson who followed closely the route of the Rochdale Canal through the Pennines. It opened section by section from 1839 to 1841. A major work was the Summit Tunnel, which took much longer to build than expected with significant increased cost. This tunnel suffered a major fire in 1984 when a train of thirteen tankers carrying over 800 tonnes of 4-star petrol caught fire in the tunnel because of a defective axle box that caused a derailment and leakage from one of the tankers. Firefighters were unable to fight the fire directly and instead blocked the ventilation shafts with foam to starve the fire of oxygen. The fire was so fierce that bricks in the wall turned to glass and melted. Nevertheless the tunnel survived, relatively minor remediation work was undertaken but the bulk of the brickwork lining the tunnel was still serviceable. George Stephenson built things to last! During construction the company was authorised to build a line from Todmorden to Burnley, and in 1850 the Burnley terminus was extended to meet the existing East Lancashire Railway line to provide a link to Blackburn and Preston.



Next came the Huddersfield & Manchester Railway, authorised in 1845 by the Huddersfield and Manchester Railways and Canal Act, and followed the route of the Huddersfield Narrow Canal. It bored through the Pennines at Standedge, alongside the existing canal tunnel, and used that tunnel to bring materials for construction and extraction of waste material. This is now the main route across the Pennines, and is currently undergoing a major upgrade – the Transpennine Route Upgrade – in which the line is being electrified and will be served by faster and more frequent services.

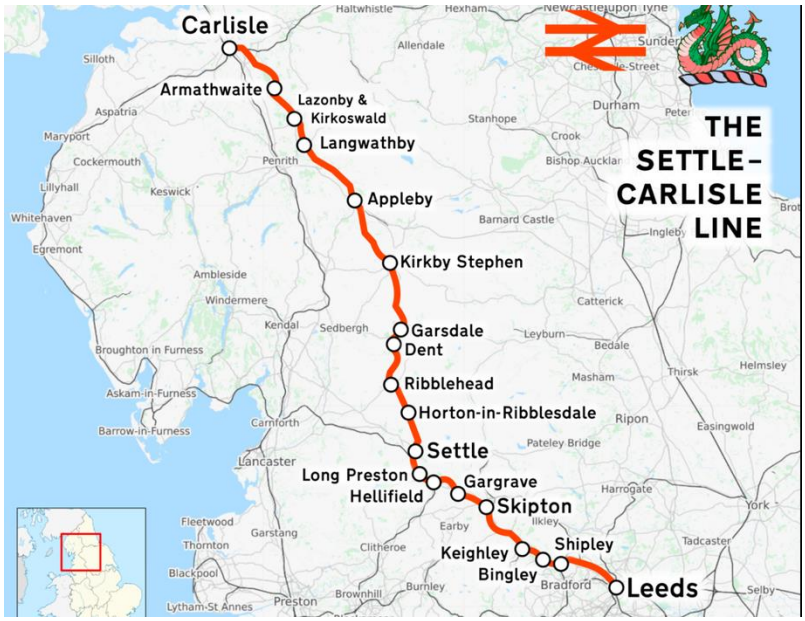


The Hope Valley Line was built by the Sheffield and Midland Railway Companies' Committee as part of the Midland Railway's drive to reach Manchester with its line from London via Ambergate and Millers Dale. Initially, in 1867, it joined the Manchester, Sheffield and Lincolnshire Railway at Hyde Junction, running into Manchester London Road (now Manchester Piccadilly). When Manchester Central Station was opened by the Cheshire Lines Committee, of which the Midland Railway was a partner, a new line was built to this station, which became the Manchester terminus for Midland trains to London St Pancras. Manchester Central Station was closed in 1969 and converted into an exhibition and conference centre, Hope Valley trains now running once again into Manchester Piccadilly via New Mills and Romiley or Stockport. There was a lack of capacity on this line with the 1985 singling of the track and removal of the island platform at Dore having caused a major bottleneck. In March 2021, it was announced that £137 million would be allocated to upgrade the line. Work started in May 2022 and was completed in April 2024. Enhancements included a new passing loop line between Bamford and Hathersage allowing passenger trains to overtake slower freight trains; the reinstatement of the second track and platform at Dore & Totley station replacing those that were removed in the 1980s. *The layout here uses the old single platform at Dore & Totley to show how single line working created such a bottleneck.*



The last and most northerly line was the Settle and Carlisle line, built by the Midland Railway after a long and bitter dispute with the London and North Western Railway over access to Scotland. In the 1860s the

Midland Railway decided that it needed its own route into Scotland and authorisation was given in 1866. Very soon after this a major banking failure occurred when interest rates rose sharply and several railways went bankrupt. The Midland had second thoughts because of the projected cost and petitioned parliament to abandon the scheme. This was refused however and construction began in 1869. The line opened to goods traffic in 1875 and passenger traffic in 1876.



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